

Sample Question Bank — Data Structures & Algorithms

Unit 1: Data Structures

1. What is a Binary Search Tree (BST)? Explain insertion and deletion operations with examples. [5 Marks]
2. Differentiate between Stack and Queue data structures with suitable diagrams. (3 Marks)
3. Which of the following data structures operates on First-In-First-Out (FIFO) principle? [1 Mark]
(A) Stack (B) Queue [Correct] (C) Binary Tree (D) Graph

Unit 2: Sorting & Searching Algorithms

4. Explain QuickSort partitioning algorithm in detail. Trace execution on [38, 27, 43, 3, 9, 82, 10] and calculate best-case time complexity. [10 Marks]
5. Define Hashing. What is a hash collision and how is it resolved using chaining? (5 Marks)